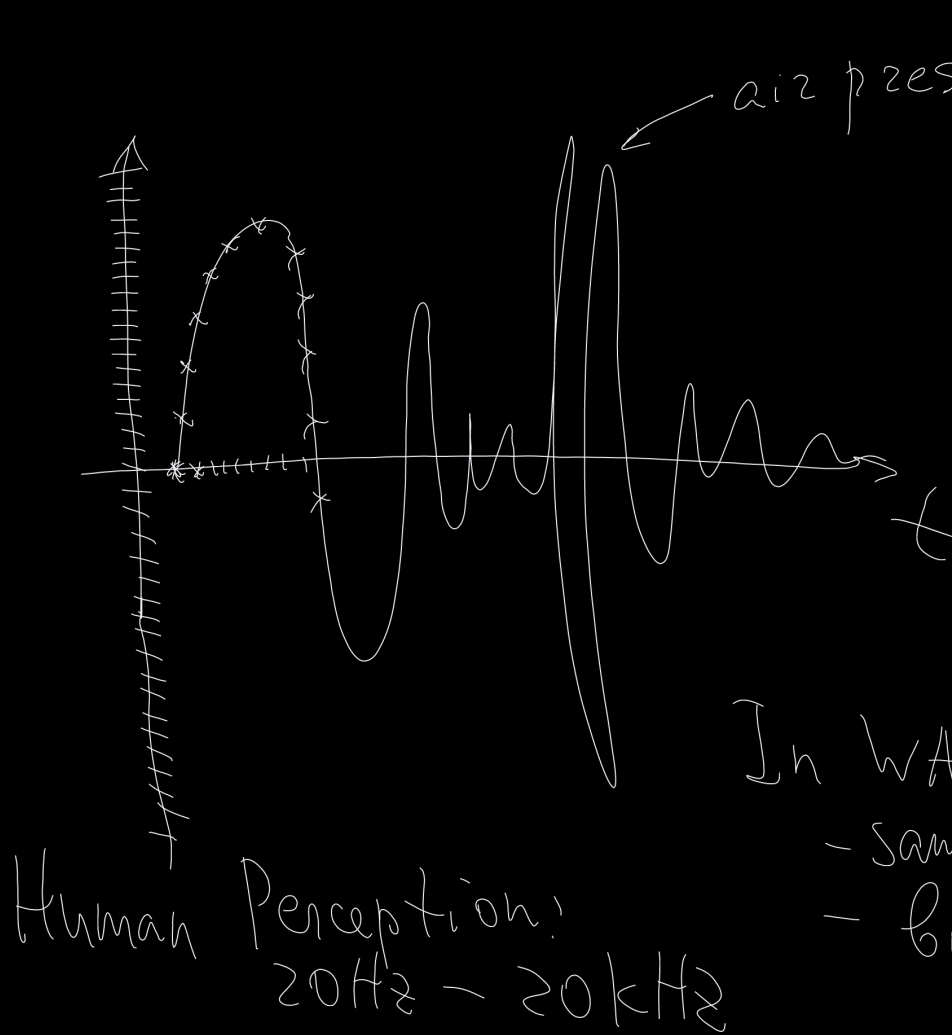


Audio Spectrograms

ASR: Speech $\rightarrow$ Spectrogram $\xrightarrow{NN}$ Text
(raw audio)

TTS: Text $\xrightarrow{NN}$ Spectrogram $\xrightarrow{NN \text{ (vocoder)}}$ Speech



air pressure (cont. signal)

digital audio

signal in computer:

- sampled in time

- finite

- quantized w.r.t. y-scale

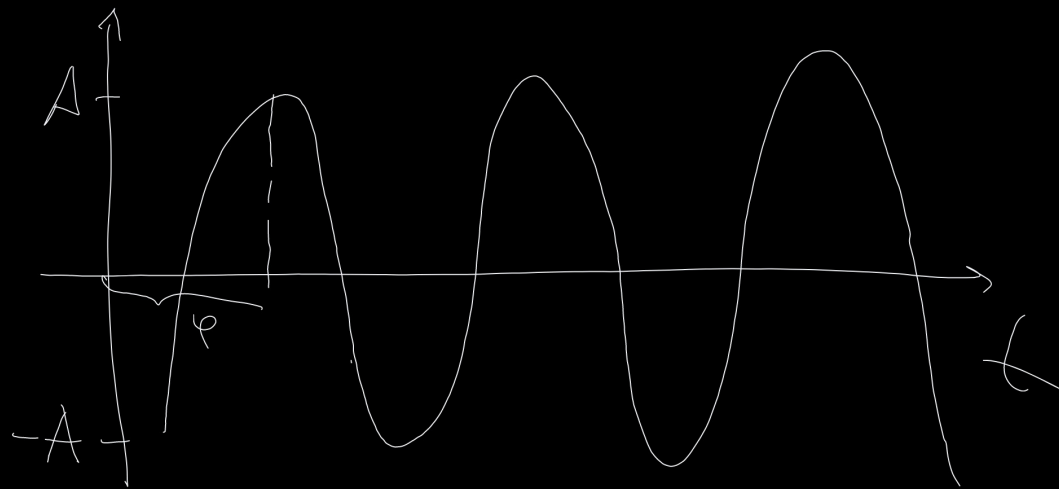
In WAV-file:

- sampling rate, e.g. 44.1kHz

- bitrate, e.g. 8-bit, 16-bit, ...

- number of channels (mono, stereo)

$$f(t) = A \cos(\omega t + \varphi)$$



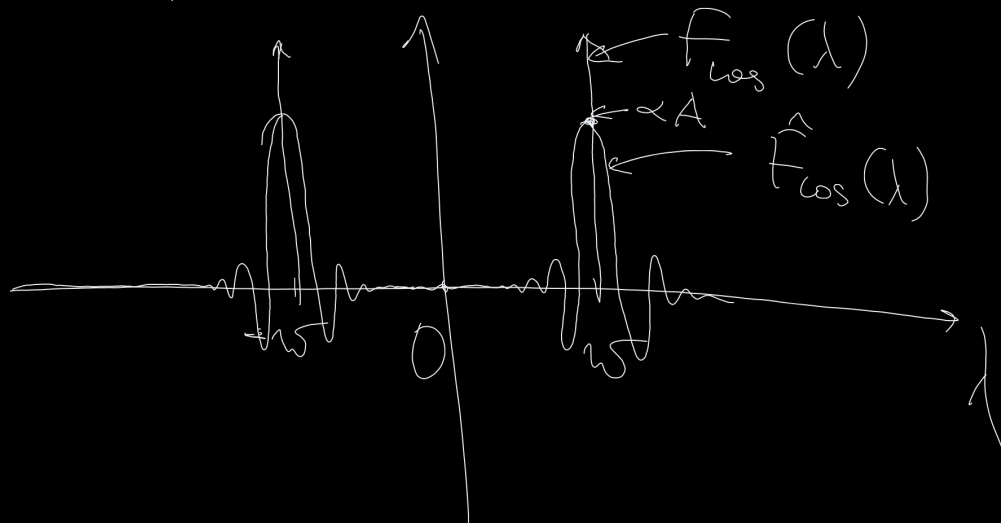
ω - frequency, Hz
 A - amplitude
 φ - phase
 pitch
 loudness

$$F_{\cos} \{ f(t) \} = A\pi (\delta(\lambda - \omega) + \delta(\lambda + \omega))$$

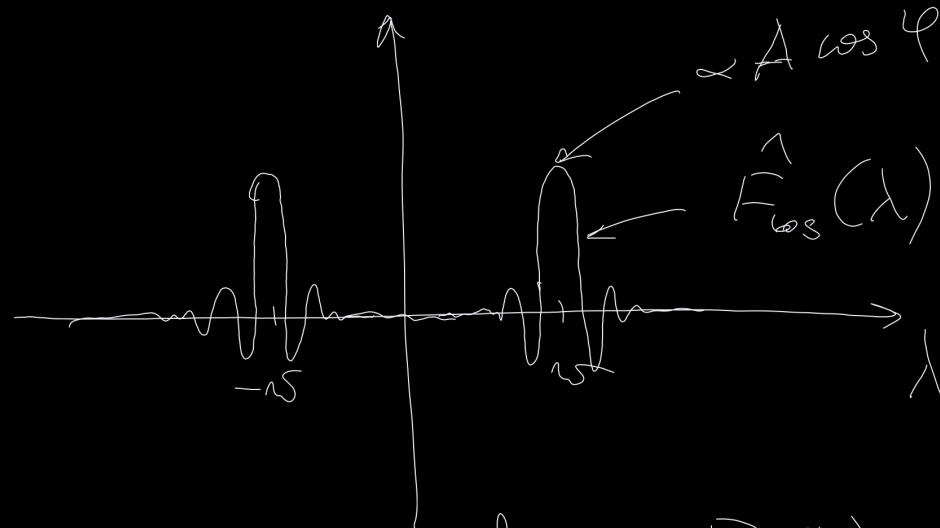
$$F_{\cos}(\lambda) = \int_{-\infty}^{\infty} f(t) \cos(\lambda t) dt \approx$$

$$f(t) = A \cos(\omega t) \approx \frac{1}{T} \int_{-T}^T f(t) \cos(\lambda t) dt \approx$$

$$\approx \frac{1}{T} \sum_{n=0}^{N-1} f\left(-T + n \frac{2T}{N}\right) \cos\left(\lambda \left(-T + n \frac{2T}{N}\right)\right) = \hat{F}_{\cos}(\lambda)$$



$$f(t) = A \cos(\omega t + \varphi)$$



Fourier Transform:

$$F(\lambda) = \int_{-\infty}^{\infty} f(t) \exp(-i\lambda t) dt$$

$\cos(\lambda t) - i \sin(\lambda t)$

$$F_{\cos}(\lambda) = \int f(t) \cos(\lambda t) dt$$

$$F_{\sin}(\lambda) = \int f(t) \sin(\lambda t) dt$$

$$|F(\lambda)|^2 = F_{\cos}^2(\lambda) + F_{\sin}^2(\lambda)$$

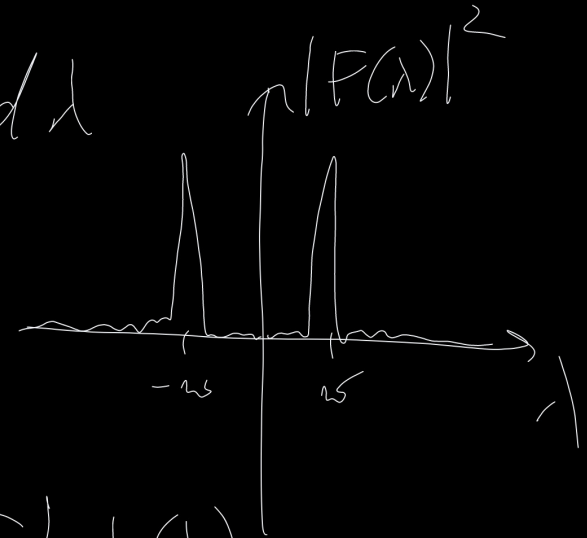
$$\varphi = \arctan\left(\frac{F_{\sin}(\lambda)}{F_{\cos}(\lambda)}\right)$$

Fourier Transform: $(f * g)(t) = \int_{-\infty}^{+\infty} f(t-\tau)g(\tau)d\tau$

- $f(t) = \frac{1}{2\pi} \int F(\lambda) \exp(i\lambda t) d\lambda$

- $|F(\lambda)|^2$ is even function

$\text{Arg } F(\lambda)$ is odd function

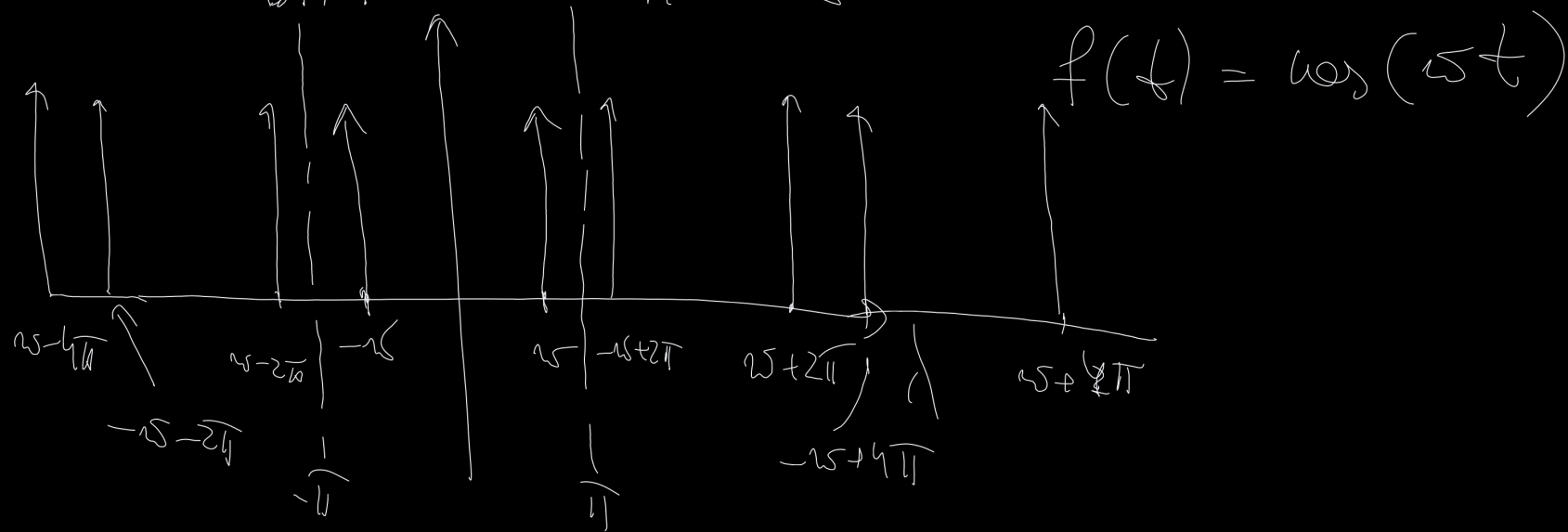


- $F\{f \cdot g\}(\lambda) = F\{f\}(\lambda) * F\{g\}(\lambda)$

$F\{f * g\}(\lambda) = F\{f\}(\lambda) \cdot F\{g\}(\lambda)$

Discrete-Time Fourier Transform

$$F_{\text{DTFT}}(\lambda) = \sum_{n=-\infty}^{+\infty} f(n) \exp(-i\lambda n)$$



$$f_1(t) = \cos(\omega t)$$

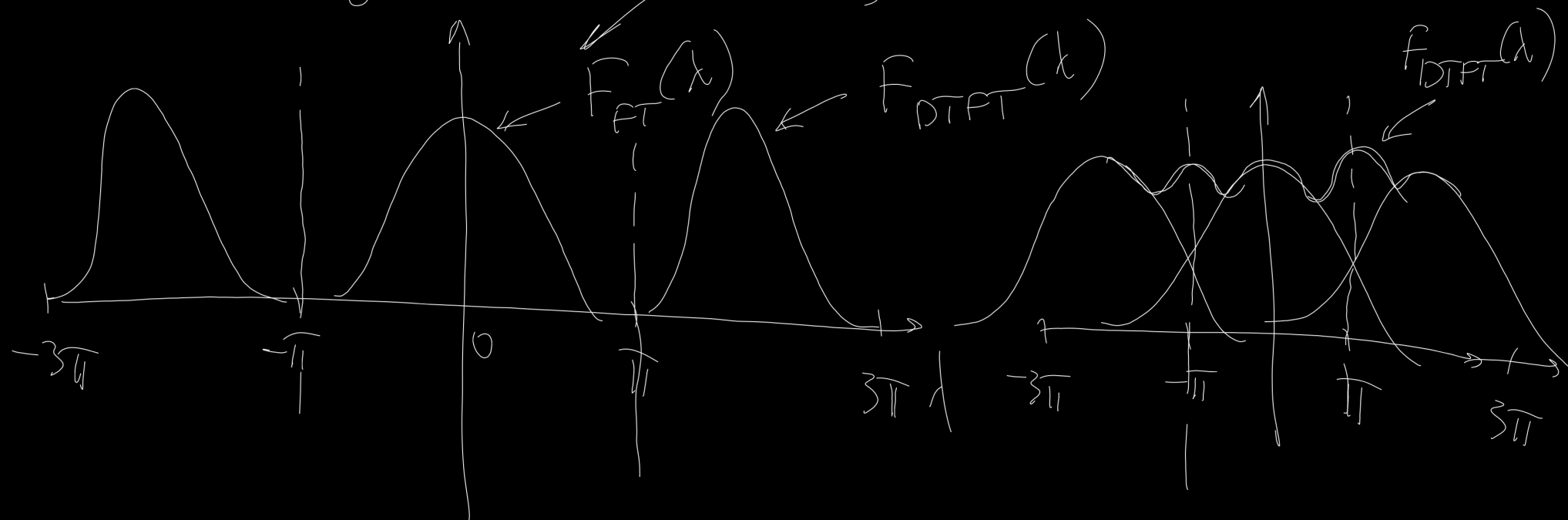
$$f_2(t) = \cos\left(\left(\omega + 2\pi k\right)t\right)$$

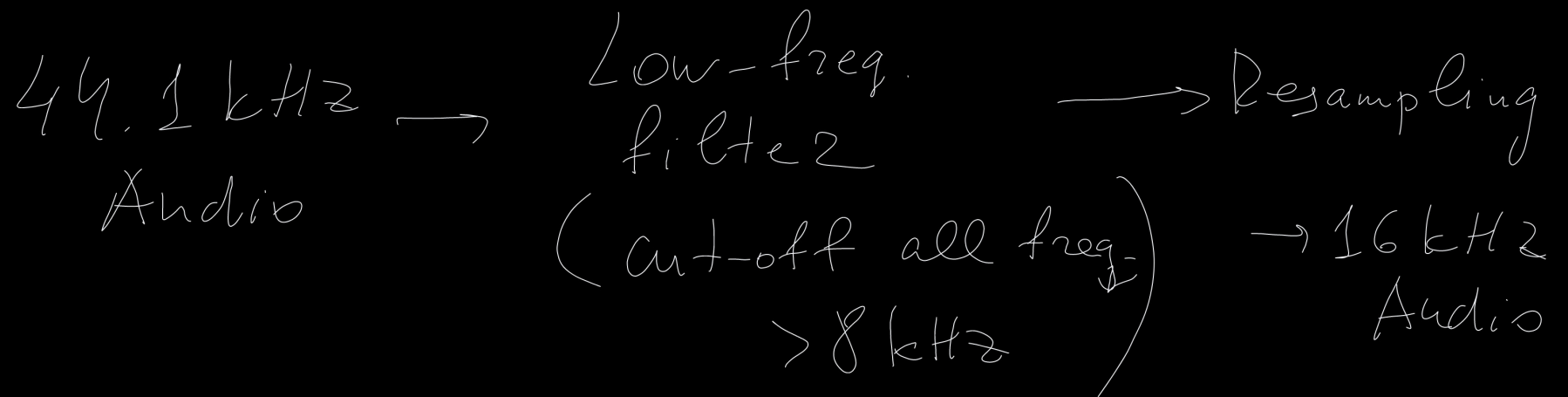
$$f_2(n) = \cos\left((\omega + 2\pi k)n\right) = \cos(\omega n) \cos(2\pi kn) - \sin(\omega n) \sin(2\pi kn) = \cos(\omega n) = f_1(n)$$

Arbitrary $f(t)$

f_s - discr. frequency, Hz

$$f_s > 2 f_{\max}$$



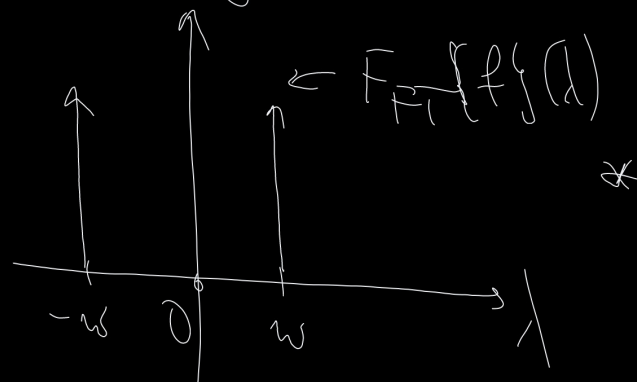


FT for finite signals

$$f(t) = \cos(\omega t), \quad g(t) = \begin{cases} 1, & \text{if } t \in [a, b] \\ 0, & \text{otherwise} \end{cases}$$

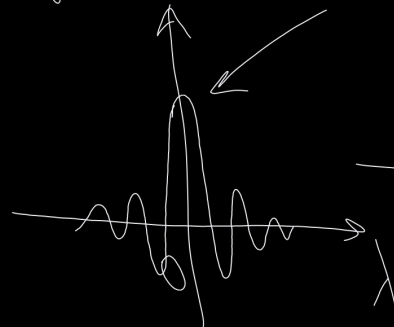
$$\hat{f}(t) = f(t)g(t)$$

$$F\{f \cdot g\}(\lambda) = F\{f\}(\lambda) * F\{g\}(\lambda)$$



$\leftarrow F_{FT}\{f\}(\lambda)$

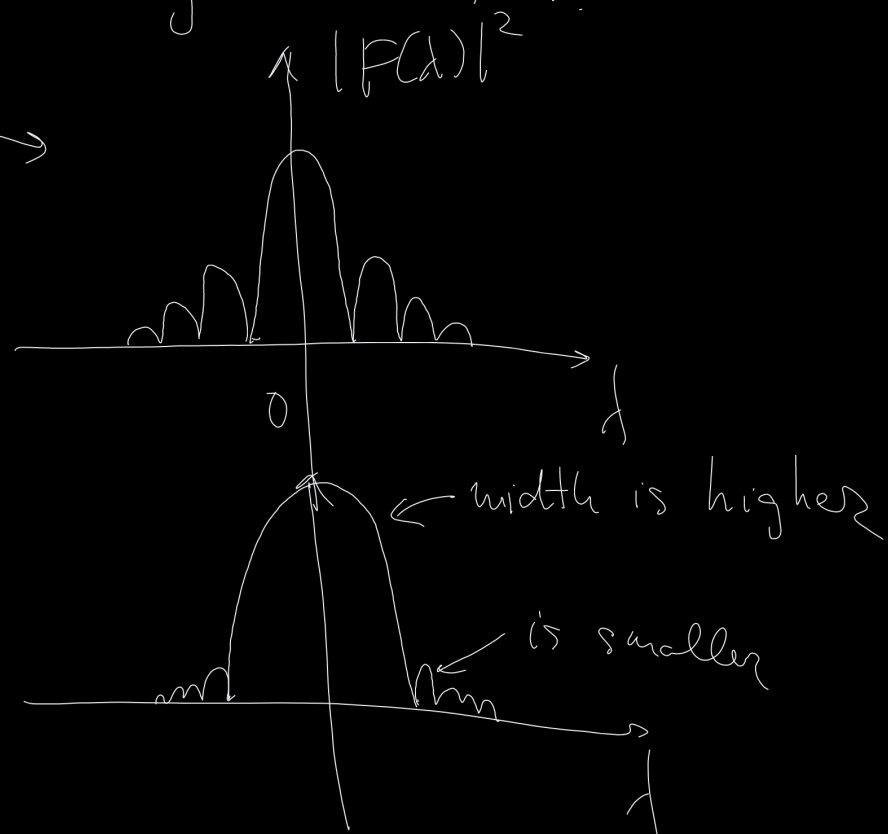
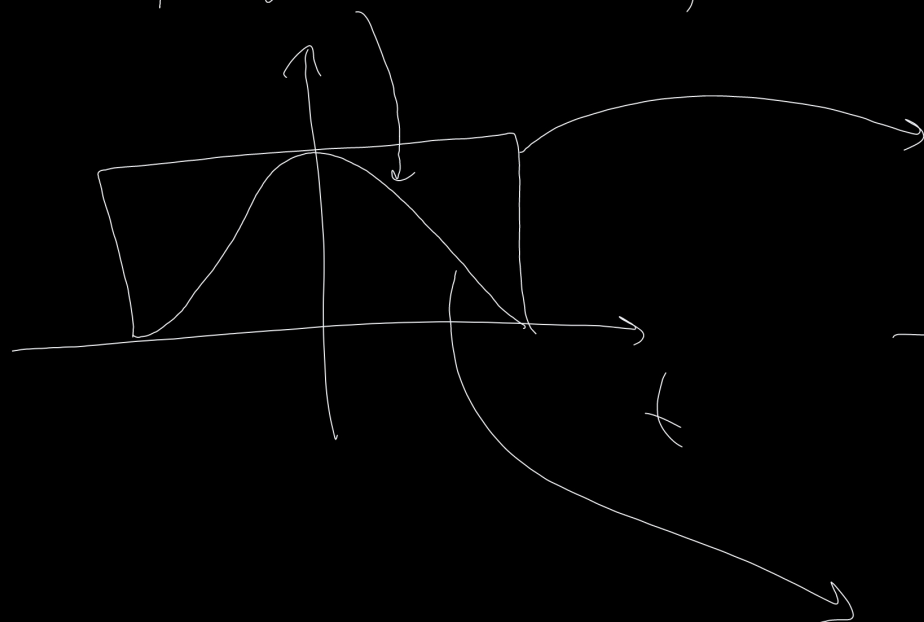
*



$\rightarrow$



flann window, flanning window, ...



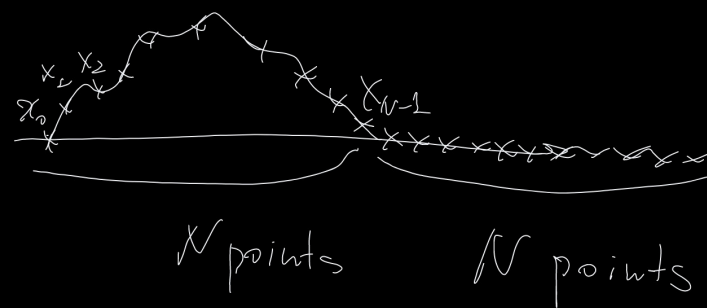
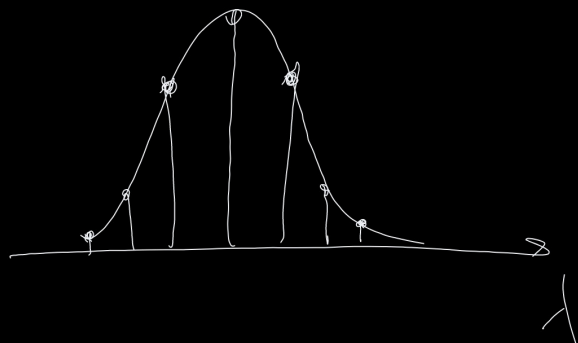
Discrete Fourier Transform (DFT)

$$x_0, x_1, \dots, x_{N-1} \xrightarrow{\text{DFT}} \underbrace{x_0}_{\uparrow \mathbb{C}}, \underbrace{x_1}_{\uparrow \mathbb{C}}, \dots, \underbrace{x_{N-1}}_{\uparrow \mathbb{C}}$$

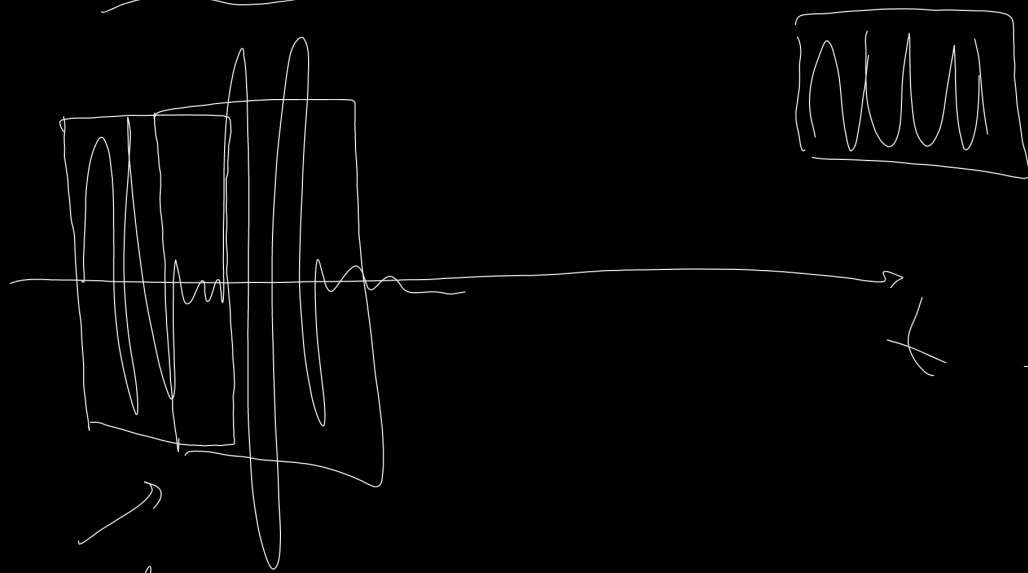
$$X_k = \sum_{n=0}^{N-1} x_n \exp(-i \frac{2\pi}{N} k n), \quad \forall k=0, 1, \dots, N-1$$

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \exp(i \frac{2\pi}{N} k n), \quad \forall n=0, \dots, N-1$$

Comp. compl.
 Direct $O(N^2)$
 FFT $O(N \log N)$



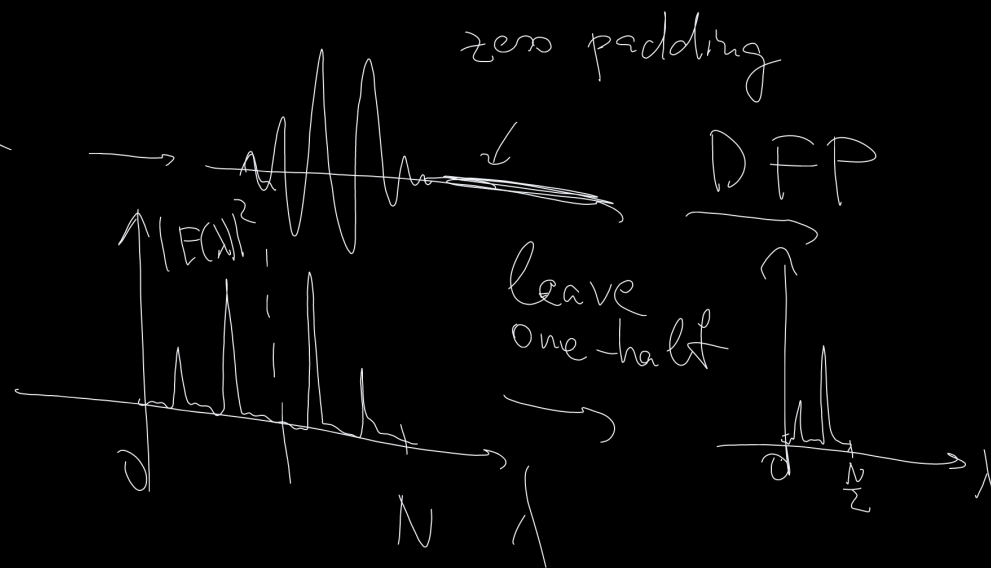
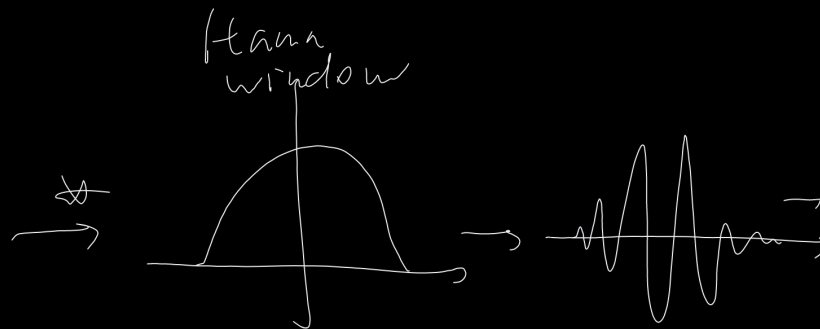
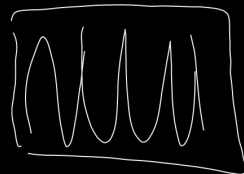
Audio Spectrogram

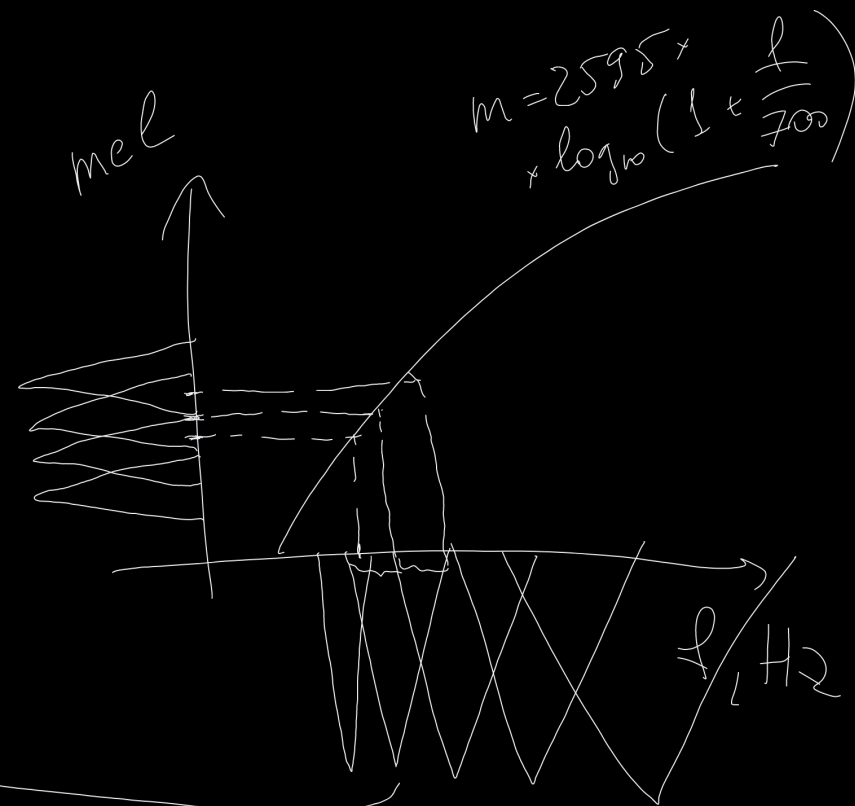
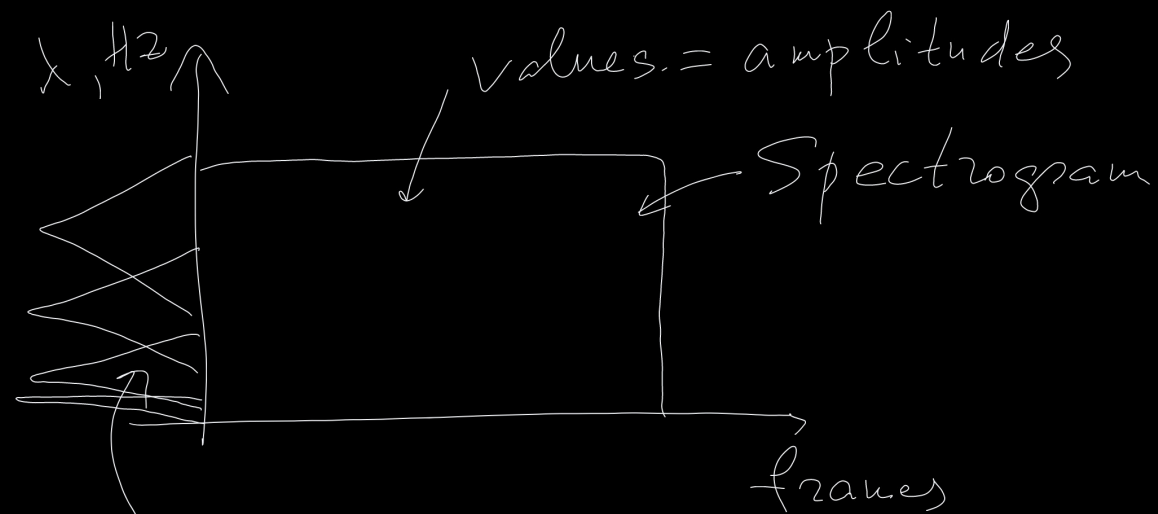


overlapping windows

~20-30 ms

75% overlap





mel-spectrogram
+ log to values = log-mel-spectrogram